LAB 3 — COMPONENT MODELLING & ARCHITECTURAL PATTERN SELECTION

Telemedicine Slot Booking & Prescription Portal

Based on Lab 1 Problem Statement #11 — Healthcare & Telemedicine

1. Scenario and Requirements

The system enables patients to discover attending physicians and specialties, view and book consultation slots, receive a secure consultation-room link, and download digitally signed prescriptions. Physicians can create and digitally sign structured prescriptions after consultation.

- Doctor and specialty discovery.
- Consultation slot viewing and booking.
- Secure consultation-room access.
- Digital prescription creation and signing.
- Patient download of signed prescription PDF.
- TLS 1.3 in transit and AES-256 at rest for protected healthcare data.
- Acceptable response latency and security performance under simulated peak load.

2. Architectural Style Analysis

Architecture	Advantages for this system	Limitations	Suitability
Layered	Clear separation of presentation, business and data concerns.	Can create tighter coupling and makes independent scaling harder.	Good
Client-Server	Centralized control and straightforward deployment.	Central server can become a bottleneck and single point of failure.	Moderate
Microservices	Independent services, fault isolation, independent scaling and deployment.	More operational and network complexity.	Best

3. Architecture Selection

We chose Microservices Architecture for the Telemedicine Slot Booking & Prescription Portal.

- Independent scaling: high-demand services such as booking and secure video access can be scaled without scaling the entire portal.
- Fault isolation and modularity: failures in one service, such as notifications or prescription generation, can be isolated from other patient-facing functions.
- Security: authentication, secure video access and prescription handling can be separated into controlled service boundaries, while protected data can be encrypted using TLS 1.3 in transit and AES-256 at rest.
- Performance: frequently used services such as doctor discovery and slot booking can be optimized and scaled independently, supporting acceptable response latency under peak load.

4. UML Component Diagram

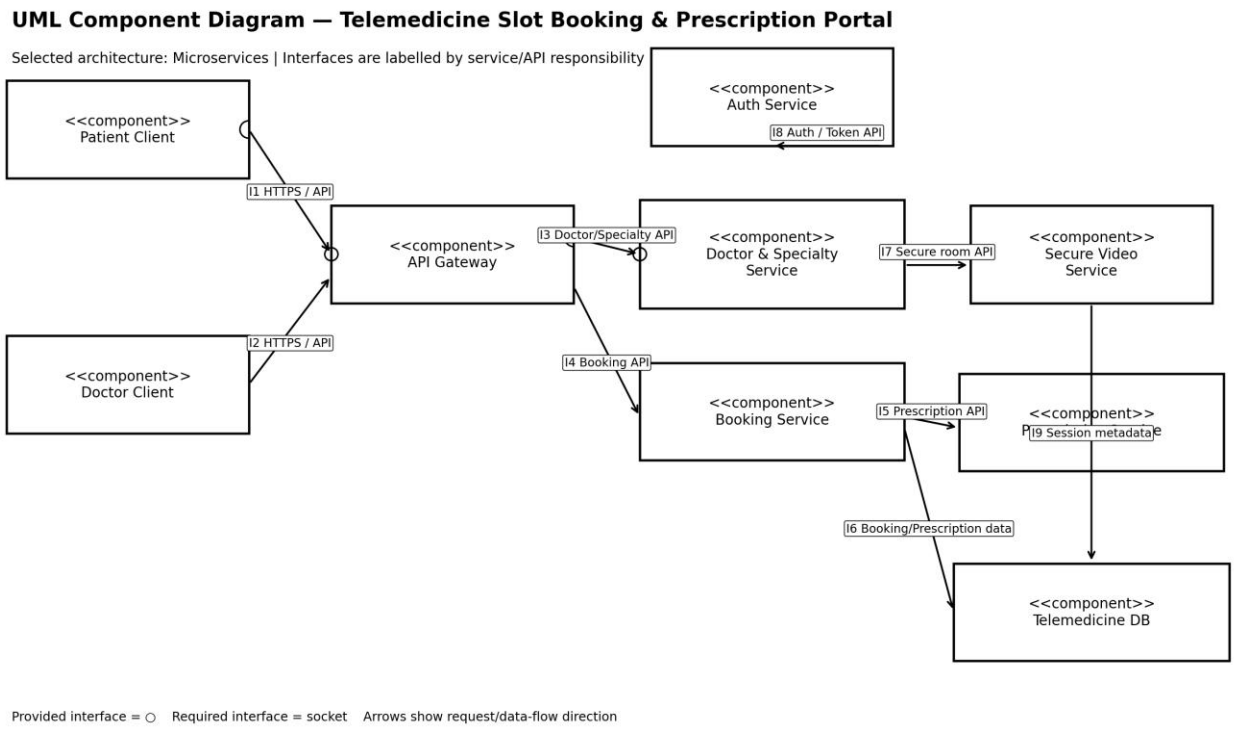


Figure 1. Component model with service interfaces and dependencies.

5. Key Component Interfaces

Interface	Connected Components	Purpose
I1	Patient Client ↔ API Gateway	Secure patient requests over HTTPS/API.
I2	Doctor Client ↔ API Gateway	Secure physician requests over HTTPS/API.
I3	API Gateway ↔ Doctor & Specialty Service	Retrieve physician and specialty information.
I4	API Gateway ↔ Booking Service	View availability and book consultation slots.
I5	Booking Service ↔ Prescription Service	Support prescription workflow after consultation.
I6	Booking/Prescription Services ↔ Telemedicine DB	Persist booking and prescription records.
I7	Booking Service ↔ Secure Video Service	Generate and manage secure consultation-room access.
I8	API Gateway ↔ Auth Service	Authenticate users and validate access tokens.
I9	Secure Video Service ↔ Telemedicine DB	Store protected session metadata.